Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		$\frac{38\,000\,000}{3\,750\,000}$ (1) Payback = 10.1 [years] (1) Accept 10 [years]		2		2	2	
		(ii)		Cost of {electricity / biomass / wood / energy} may change **OR** Energy use may change OWTTE Accept bills could change or savings could change		1		1		
	(b)			Growing trees (1) {removes / absorbs} CO_2 from the atmosphere (1) [so disagree] **Alternative:** A tree absorbs as much CO_2 (1) as it releases (1)			2	2		
	(c)	(i)		3 + 13.4 = 16.4 [MW] **or** 20.0 **–** 3.6 = 16.4 [MW]		1		1	1	
		(ii)		Efficiency = $\frac{16.4\ \textbf{ecf}}{20.0} \times 100$ (1) Efficiency = 82% (1) so disagree Conclusion must be present to award both marks **Alternative:** $\frac{15}{100}$ x 20 (1) Useful output = 3 M[W] (1) so disagree Conclusion must be present to award both marks			2	2	2	

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(d)	(i)		Substitution: energy transfer = power $\times$ time i.e. 20×60 (1) Conversion i.e. 20×3600 (1) Energy transfer = 72 000 [MJ] (1) Award 2 marks for an answer of 1200 [MJ] Award 2 marks for an answer of 7.2×10^{n} except for when $n \neq 4$ Award 1 mark for an answer of 1.2×10^{n} except for when $n \neq 3$	1	1 1		3	3	
		(ii)		$\frac{72\,000\ \textbf{ecf}}{2880} = 25$ [tonnes]		1		1	1	
		(iii)		Substitution: volume = $\frac{25\ \textbf{ecf} \times 1000}{500}$ (1) Volume = 50 [m^3] Award 1 mark for answer of 0.05 [m^3]	1	1		2	2	
		(iv)		$\frac{50\ \textbf{ecf}}{5} = 10$		1		1	1	
				Question 7 total	**2**	**9**	**4**	**15**	**12**	**0**